

The inferior colliculi are hemispherical and somewhat more prominent than the superior [MidBrain Activation](#) . CN III is the oculomotor nerve. The spinal nerve is formed by the joining of the dorsal and the ventral roots. The damage is irreversible and treatment is centered on rehabilitation and preventing another stroke. The front and bottom areas of the temporal lobe are involved in smell and functions of the limbic system. Fraser however denies the existence of such fibers. It controls muscles of the pharynx larynx and velum. It coordinates subconscious movements such as balance and coordinated movement. The main arterial blood supplies to the brain. Just reading this text now you are using your [Midbrain](#) to keep your eyes and head moving.

It is separated from the inner pia layer by subarachnoid space which is filled with cerebrospinal fluid. The origin and course of the superior peduncle have already been described . Many fibers either terminate in or send off collaterals to the gray matter of the medulla the pons and the mid-brain. From an evolutionary perspective the hindbrain is the oldest part of our brain and is located deep within our head and on top of our spinal cord. The postcentral gyrus comprises the primary somatosensory cortex which is concerned with somatosensory reception integration and processing sensory information from the surface of the body and from the viscera and is important for the formulation of perception. The temporal lobe controls the hearing centers language comprehension storing and recalling memories and related areas including some speech centers. long situated between the corpora quadrigemina and tegmenta and connecting the third with the fourth ventricle [MidBrain Activation For Adults](#) . We will keep to the use of the term Midbrain as this has been popularised by recent developments. Besides these scattered cells it contains three groups which constitute the nuclei of the oculomotor and trochlear nerves and the nucleus of the mesencephalic root of the trigeminal nerve. Portions of the [Midbrain](#) called the red nucleus and the substantia nigra are involved in the control of body movement.

It is located deep in the temporal lobe. Projections from the axons of these pontine neurons form large transverse fiber bundles that traverse the pons and ascend to the contralateral cerebellum via the middle cerebellar peduncles. The brain stem is made of the [Midbrain](#) pons and medulla [MidBrain Activation For Adults](#) . Caudal to the postcentral gyrus is the inferior parietal gyrus. The primary function of the ANS is to regulate and control unconsciousness functions including visceral smooth muscle cardiac muscle vessels and glandular function . The surface of the cerebrum is made up of gyri and sulci. The caudal part of the superior temporal gyrus which extends up to the parietal cortex forms part of Wernickes area [MidBrain Activation](#) . The forebrain consists of the cerebral cortex the basal ganglia the limbic system the thalamus and the hypothalamus. Not all structures are mentioned but having watched these you might be able to invent your own memory links [MidBrain Activation](#) . Lateral to the putamen is the insula while more dorsomedially the corpus callosum overlies the cavities of the lateral ventricles.

It descends in the adjoining region of the ventral and lateral funiculi [MidBrain Activation](#) . Additionally facial paralysis due to a pyramidal lesion will not permit voluntary control of the muscles but these paralyzed muscles will respond to emotional expression. The cerebral cortex consists of various types of cortices as well as closely related subcortical structures such as the caudate nucleus putamen globus amygdala and the hippocampal formation . It is involved in memory attention language thinking and voluntary movement. Transverse section of mid-brain at level of inferior colliculi. For this it is necessary to activate the neighbouring pineal body. The [Midbrain](#) helps coordinate sensory information with simple movements. The upper surface of the superior temporal gyrus which extends into the lateral fissure is called the transverse temporal gyrus and is the primary auditory cortex. The orbital portion of the prefrontal cortex is important in the appropriate switching between mental sets and the regulation of emotion. The efferent projections include cortico bulbar cortex to brain stem and cortico spinal cortex to spinal cord axons.

Also included in the PNS are the sensory receptor organs. Its posterior border is separated from the cerebellum by the aqueduct of Sylvius and more inferiorly by the fourth ventricle. This progressive illness develops when dopamine-producing nerve cells in the pars compacta die off in large numbers.

It is thought that the neocortex is a recently evolved structure and is associated with higher information processing by more fully evolved animals such as humans primates dolphins etc. The ascending reticular formation is also called the reticular activating system. The pineal body secretes two hormones: melatonin and serotonin. The **Midbrain** tectum is formed by two pairs of rounded structures: the superior and inferior colliculi. The cerebral cortex is the part of the brain that functions to make human beings unique. The circle of Willis ensures a continuous blood supply to the brain. In general the CNS can be divided into three main functional components: the sensory system the motor system and homeostatic and higher brain functions.

These convolutions allow for the expansion of the cortical surface area without increasing the size of the brain. This means we end up using much less of our brain than we actually could. The process of activating the **Midbrain** reverses this trend and allows us to use our brain more efficiently hence the improvement in cognitive abilities. The medulla is located directly above the spinal cord and controls many vital autonomic functions such as heart rate breathing and blood pressure **MidBrain Activation For Adults**. Each bump on the surface of the brain is known as a gyrus while each groove is known as a sulcus. The substantia nigra black substance is so called because of the higher levels of melanin in its dopaminergic neurons. The Limbic System is sometimes called the emotional control center of our brain because it contains structures that help us feel our most raw emotions **MidBrain Activation For Adults**. Also within the pons base and tegmentum are longitudinally ascending and descending fibers. In the following descriptions the terms are applied to a standing human. The metencephalon and myelencephalon together compose the rhombencephalon which divides into pons and medulla oblongata. This part of the reticular system projects to the mid-line group of the thalamus which also plays a role in wakefulness.

The superior and inferior colliculi are involved in visual and auditory functions respectively. The mesencephalon consists of several structures around the cerebral aqueduct such as the periaqueductal gray the mesencephalic reticular formation the substantia nigra the red nucleus the superior and inferior colliculi the cerebral peduncles some cranial nerve nuclei and the projection of sensory and motor pathways. They begin and end within the brain stem. The insula lies in the depths of the lateral sulcus. The surface of the cerebral cortex is highly convoluted with folds separate from each other by elongated grooves. Dopamine is a neurotransmitter a type of chemical that is essential for the movement of electrical signals between brain cells. It is involved in the coordination of motor movements as well as basic facets of memory and learning. It is also an important part of the extra pyramidal tract. The rubrospinal tract forms an important part of the pathway from the cerebellum to the lower motor centers **MidBrain Activation**. The terms rostral and anterior refer to a direction towards the face.